

CampusEats — External SOAP Partner Integration

CS543 Web Services — Assignment 3

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1. Context

CampusEats integrates its Payments Service with an external payment gateway called UniPay using SOAP. The operation used for this integration is **charge**. SOAP is suitable at this partner edge because the external gateway provides a formal WSDL contract and supports structured payment messages. The rest of CampusEats can remain REST-oriented while this external payment edge uses SOAP.

2. HTTP Binding

The SOAP request is carried by HTTP POST using the endpoint and SOAPAction defined by the partner WSDL.

```
POST /pay HTTP/1.1
Host: api.unipay.example
Content-Type: text/xml; charset=utf-8
SOAPAction: "urn:unipay:charge"
```

Endpoint: `https://api.unipay.example/pay`

The request body contains a SOAP Envelope with authentication credentials in the Header and the charge operation in the Body.

3. Discovery

CampusEats would obtain the UniPay WSDL through a service catalogue/registry or a direct WSDL URL supplied by the partner. No live UDDI server is required.

Registry Entry

Business	UniPay
Service	Payments
Interface	SOAP / WSDL
Endpoint	<code>https://api.unipay.example/pay</code>
WSDL	<code>https://api.unipay.example/pay?wsdl</code>
Operation	charge
tModel-style pointer	urn:unipay:PaymentsWSDL

4. Fault Mapping

The external UniPay SOAP fault **card_declined** is mapped to the existing CampusEats contract error **payment declined**.

Partner fault: card_declined

CampusEats error: payment declined

The partner-specific fault code is handled inside the Payments integration and is not exposed to CampusEats students or callers.

5. Integration Flow

1. A student places an order in CampusEats.
2. The Orders service requires payment.
3. The Payments service prepares a charge request.
4. The request is sent to UniPay using SOAP over HTTPS.
5. UniPay processes the payment.
6. UniPay returns either a successful charge response or a SOAP Fault.
7. On success, CampusEats receives the transaction ID and payment status.
8. If the card is declined, card_declined is mapped to payment declined.

6. Supporting Files

The required companion artifacts are:

1. partner.wsdl
2. soap-request.xml
3. soap-response.xml
4. soap-fault.xml

This PDF documents the SOAP partner integration, HTTP binding, discovery, registry information, and fault mapping.